

Syllabus for

Academic year 2020/2021 >

TIN214 - Computer programming ⚠

Programmeringsteknik

Syllabus adopted 2020-02-19 by Head of Programme (or corresponding)

Owner: [TKTFY](#)

6,0 Credits

Grading: TH - Pass with distinction (5), Pass with credit (4), Pass (3), Fail

Education cycle: First-cycle

Major subject: Computer Science and Engineering, Engineering Physics

Department: 37 - COMPUTER SCIENCE AND ENGINEERING

Teaching language: Swedish

Application code: 57147

Open for exchange students: No

Only students with the course round in the programme plan

Credit distribution

Module	Sp1	Sp2	Sp3	Sp4	Summer course ⓘ	No Sp	Examination dates ⓘ
0120 Laboratory 3,0 c Grading: UG		3,0 c					
0220 Examination 3,0 c Grading: TH			3,0 c				Contact examiner, Tue 08/06-2021 pm J ⚠ Tue 17/08-2021 am J ⚠

In programs

[TKTEM ENGINEERING MATHEMATICS, Year 1 \(compulsory\)](#)

[TKTFY ENGINEERING PHYSICS, Year 1 \(compulsory\)](#)

Examiner:

[Aarne Ranta](#)

 [Go to course homepage](#)

Eligibility

General entry requirements for bachelor's level (first cycle)

Applicants enrolled in a programme at Chalmers where the course is included in the study programme are exempted from fulfilling the requirements above.

Specific entry requirements

The same as for the programme that owns the course.

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Aim

Computer programming is a basic skill needed in both studies and professional life. This course gives the basic skills that the participants can both accomplish programming tasks typical of science and engineering tasks and understand the fundamental principles of computers and programming.

Learning outcomes (after completion of the course the student should be able to)

Knowledge and understanding

- Express mathematical formulas as programming language expressions and algorithms
- Choose appropriate datatypes and datastructures for different kinds of data
- Structure large programs into manageable and reusable units by the use of concepts such as modules, classes, and functions
- Search and find relevant program libraries and use them in adequate ways
- Explain the relations between hardware, operating system, and user programs
- Use programming for basic data analysis involving large files

Competence and skills

- Write programs that manipulate numeric and textual data to perform standard programming tasks
- Build interactive programs with both text-based and graphical user interfaces
- Make programs that read, transform, and generate files in the file system
- Use standard libraries and follow best programming practices
- Test programs by methods such as unit, regression, and property-based testing
- Use programming tools such as code editors and revision control systems

Judgement and approach

- Assess the difficulty and resources needed for typical programming tasks
- Analyse code written by others and find errors and possibilities for improvement

Content

The course is a first introduction to programming by using a general-purpose programming language. It gives a comprehensive knowledge of the language, enabling the student to write code for a wide variety of tasks and to read and reuse code written by other programmers. In parallel to learning the language, the student will be introduced to the basic concepts of computers and programming. These concepts include:

- Datatypes and their binary representations in the computer
- The relation between source code, compiler, and the machine
- The syntactic structure of expressions, statements, functions, classes, modules, etc
- The basics of algorithmic design by the use of iteration, recursion, decomposition to functions, and top-down vs. bottom-up
- The basics of software design in terms of functions, classes, modules, etc

Organisation

The course consists of lectures and exercise sessions. There is also a number of programming assignments where the skills can be trained in practice

Literature

Information about literature will be given on the course home-page before the start of the course

Examination including compulsory elements

Both modules include mandatory programming assignments. Course grade is based on a written exam, provided all mandatory assignments have been passed.